

What can we do with Data?



Learning Outcome

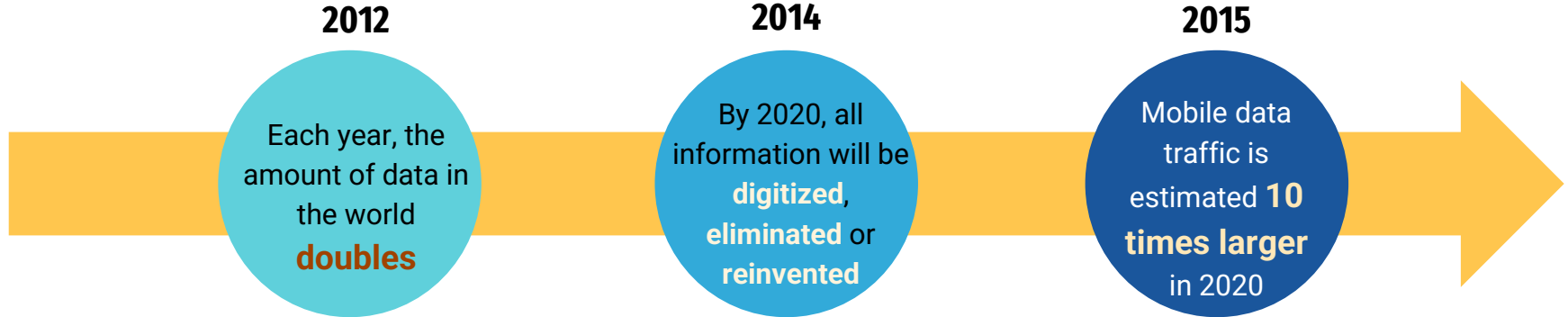
By the end of this lesson, the students will be able to

- Explain what can we do with data
- Differentiate between big data and small data
- Understand the taxonomy of data analytics
- Give examples of data usage



Data Explosion

- Data previously produced on paper, but now it's moving **online**
- **Source of Data** : Web, social media, machines, sensors
- Benefit : Rich source of information that **can be transformed** into **useful**, **valid**, and **human-understandable knowledge**



Big Data Statistics

Twitter

Twitter users send over **half a million tweets every minute**



Time

Internet users spent a total of **1.2 billion years online**



Data Interaction

went up by 5000% between 2010 and 2020



WhatsApp

WhatsApp users exchange up to **65 billion messages daily**



Google

3.5 billion searches daily



Person

generated **1.7 megabytes per second**



Source : <https://techjury.net/blog/big-data-statistics/#gref>

Data Analytics



Definition

The science that analyze crude data to extract useful knowledge from them

Process

Data collection, organisation, pre-processing, transformation, modeling, interpretation



Source : <https://www.digitaltechconsultancy.com/big-data>

Term to describe the exponential growth, availability, and use of information
(structured/unstructured)

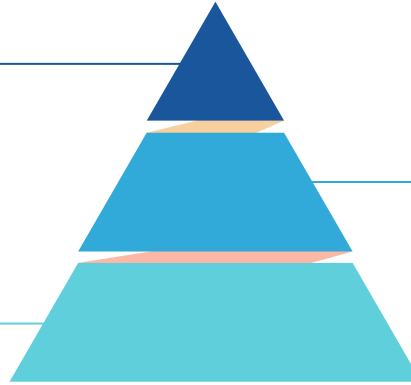
V's of Big Data

Volume

Refers to the **size** of the data` obtained, which is the most essential and distinctive feature of Big Data

Velocity

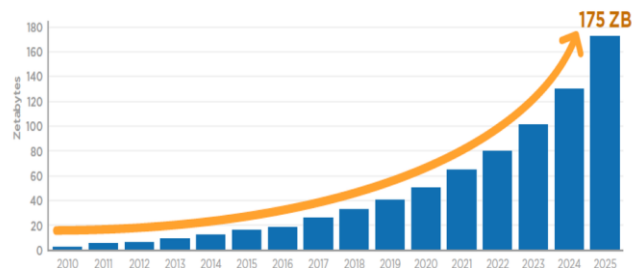
Intends to measure **how quickly** the data are processed from the creation until the visualisation of the data



Variety

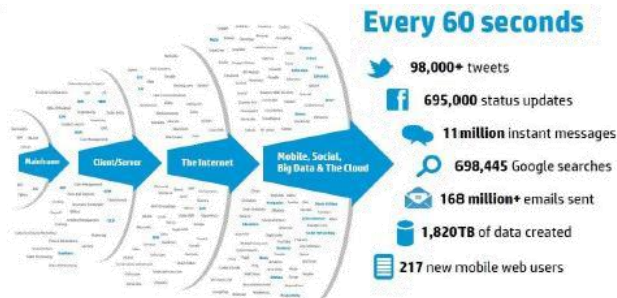
Refers to the **different types of data** that can be processed

Big Data Architecture



Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018

<https://medium.com/analytics-vidhya/the-5-vs-of-big-data-2758bfcc51d>

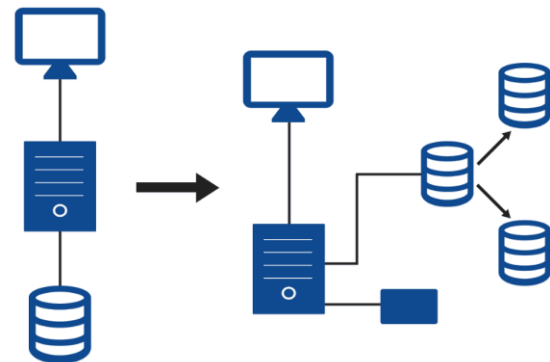


https://www.researchgate.net/figure/Examples-of-big-data-velocity_fig3_313400371



<https://www.analyticsvidhya.com/blog/2020/11/what-is-big-data-a-quick-introduction-for-analytics-and-data-engineering-beginners/>

Data increase in
volume, velocity and
variety



<https://medium.com/swlh/5-database-scaling-solutions-you-need-to-know-e307570efb72>

Scalability

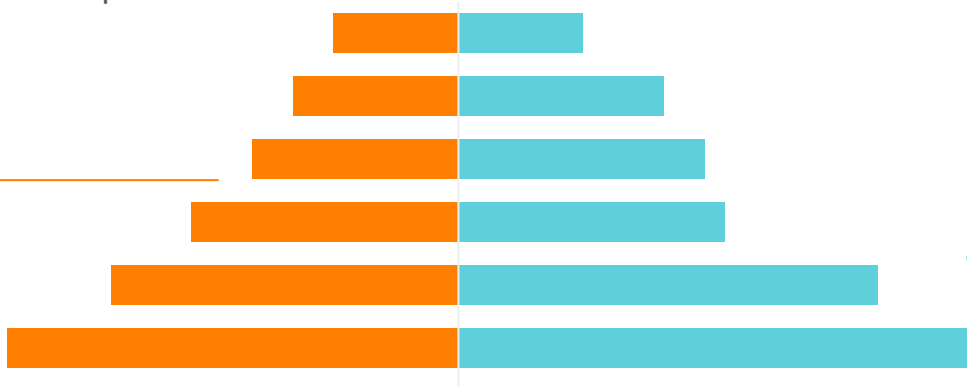
MapReduce

WHAT?

- Method of programming – way of thinking (**not programming language**)
- Originally created by Google
- Design to solve one problem
- Map – Reduce

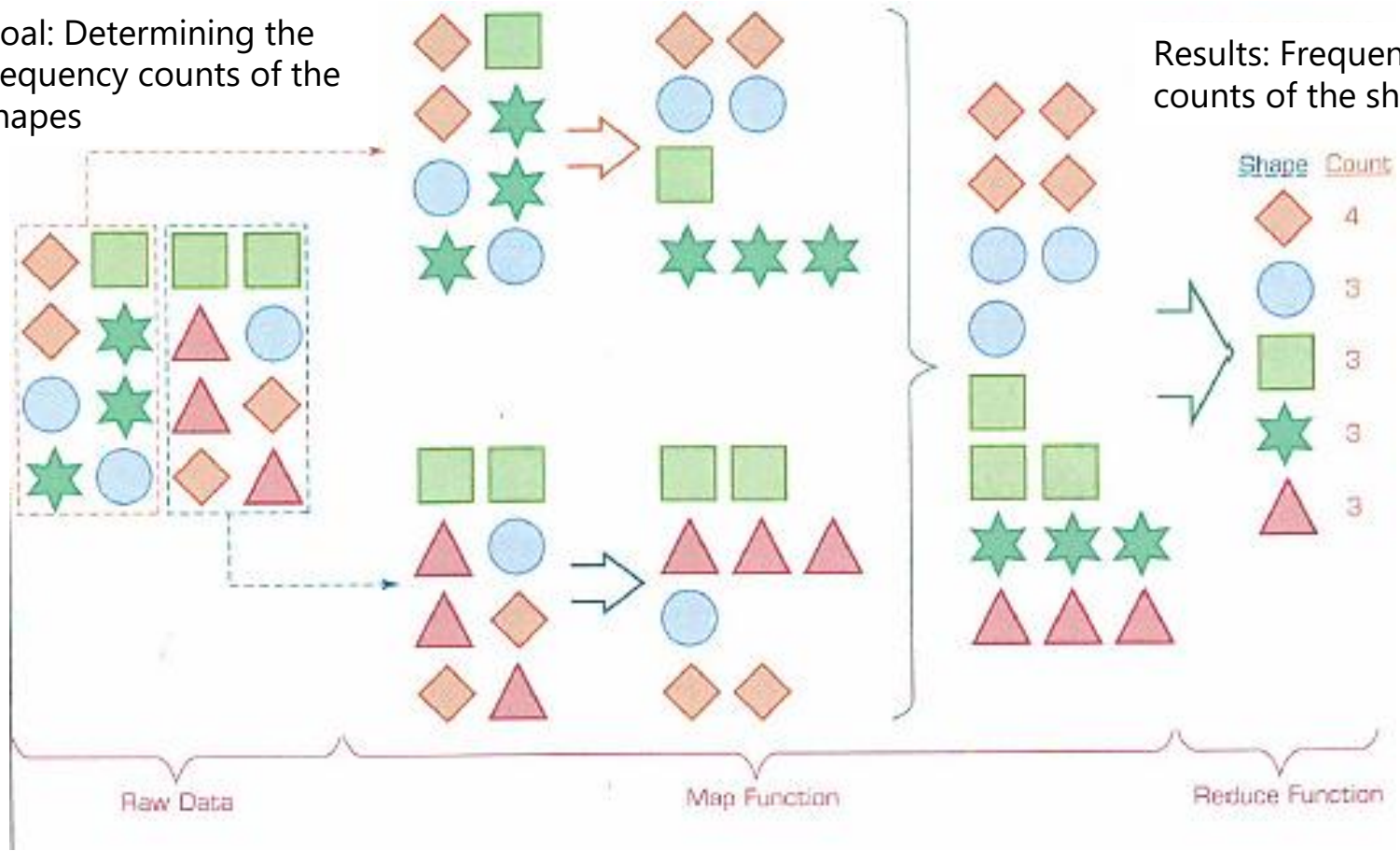
WHY?

Able to tackle **huge data** sets by **distributing processing** across many nodes, and then **combining** or **reducing** the results of those nodes.



Goal: Determining the frequency counts of the shapes

Results: Frequency counts of the shapes



Small Data

Dataset whose **volume** and **format** allows it processing and analysing by a **person/ small organisation**

Can be created by different people in distributed and integrated way

When **small data** collected to be stored and processed in large data = **BIG DATA**

Small Data vs Big Data

BIG
DATA
tells us
what's
happening.



small data tells us
why it's
happening.

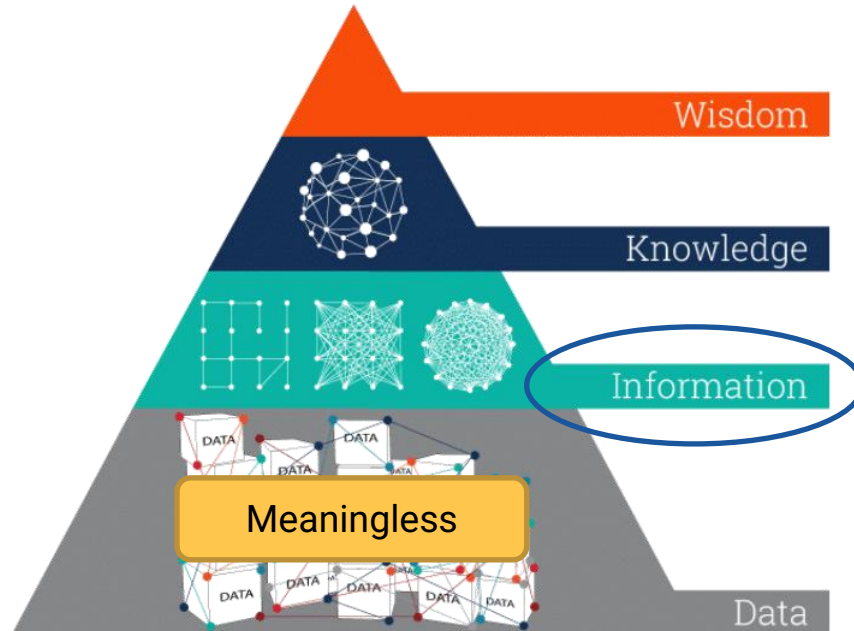


What is DATA?



What is Data?

- **Data** - a large set of **bits encoding numbers, texts, images, sounds, videos,** and so on.



Attribute vs Instance

Instance

Examples of the
concept we want
to characterize

Contact	Age	Educational level	Company
Andrew	55	1.0	Good
Bernhard	43	2.0	Good
Carolina	37	5.0	Bad
Dennis	82	3.0	Good
Eve	23	3.2	Bad
Fred	46	5.0	Good
Gwyneth	38	4.2	Bad
Hayden	50	4.0	Bad
Irene	29	4.5	Bad
James	42	4.1	Good
Kevin	35	4.5	Bad
Lea	38	2.5	Good
Marcus	31	4.8	Bad
Nigel	71	2.3	Good

Attribute

Characteristics of
an instance

Dataset

- Data sets represented by several tables
- Relations between these tables - relational data sets
- This information is easily handled using relational databases

Name	Dry/Wet Food	Good Boy (Y/N)
Fido	Dry	Y
Rex	Wet	N
Bubbles	Dry	Y
Cujo	Wet	N

Tag #	Height (in)	Weight (lbs)
1573	15	21
2684	9	7
3795	27	130
4806	6	5

Tag #	Name	Breed	Color	Age
1573	Fido	Beagle	Brown/White	1.5
2684	Rex	Pekingese	White	9
3795	Bubbles	Rottweiler	Black	5
4806	Cujo	Chihuahua	Gold	4

Taxonomy of Data Analytics

Predictive analytics/ Supervised Learning

- To **predict the value** of attribute based on value of another attribute
- Result obtained directly by applying an algorithm to the data.
- E.g: **Statistic**, such as an average, a plot, or a set of groups with similar instance

Descriptive analytics/ Unsupervised Learning

- Extract **models** from data to be used for future predictions
- Exploratory in nature
- Require post-processing technique to validate result

Supervised Learning

01



Classification

- Discrete variable
- Predict whether user will make a purchase at bookstore



02



Regression

- Continuous value
- Forecast future stock price

Source:
<https://mortgageadvisorslancaster.co.uk/when-did-house-prices-increase-and-decrease-the-fastest-in-the-uk-in-2017/>

Unsupervised Learning

01

Association analysis

- To discover patterns that describe associated features in data

02

Cluster analysis

- Find groups of closely related observations
- Object within group similar to one another

03

Anomaly detection

- Identify observation whose characteristics are significantly different from the rest of the data
- Also called as **Outlier**

Definition of Terms



Method/Technique

A systematic procedure that allows us to achieve an intended goal



Algorithm

- Self-contained, step-by-step **set of instructions** easily **understandable** by humans, allowing the implementation of a given method.
- Easily translated to an arbitrary programming language.



Model

- A **generalization obtained from data** that can be used afterwards to generate predictions
- A **prototype** that can be used to make predictions.

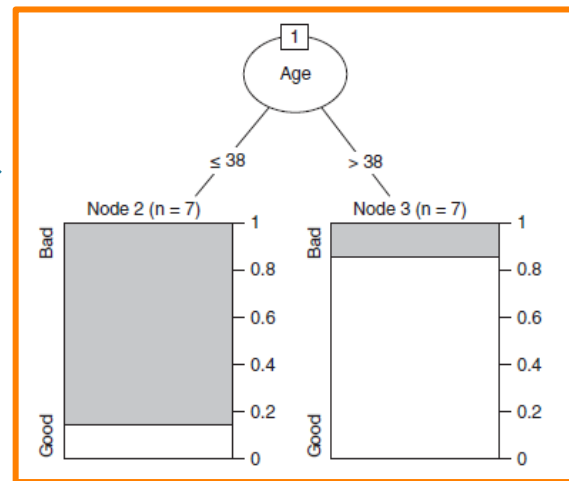
Method > Algorithm > Model

The method to **obtain the average age** of my contacts **uses the ages of each**



Algorithm An algorithm to calculate the average age of our contacts

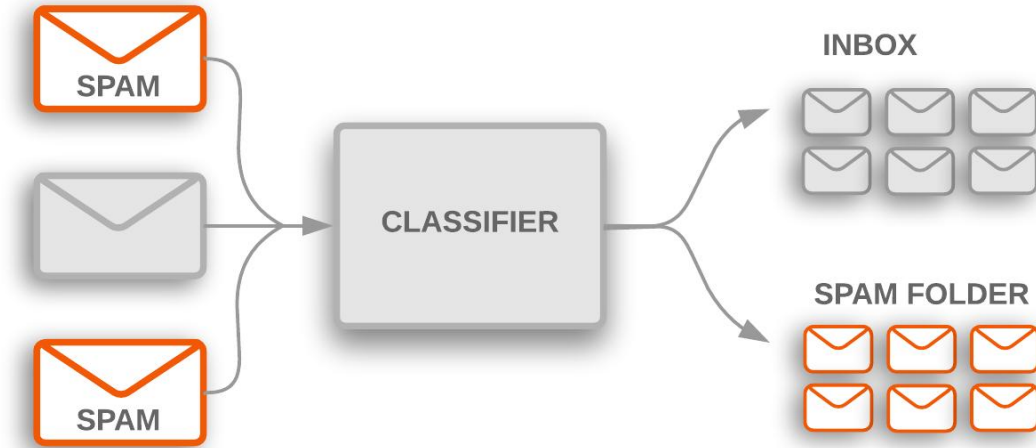
- 1: INPUT: A : a vector of size N with the ages of all contacts.
- 2: $S \leftarrow 0$ ▷ Initialize the sum S to zero
- 3: **for** $i = 1$ **to** N **do** ▷ Iterate through all the elements of A .
- 4: $S \leftarrow S + A_i$ ▷ Add the current (i th) element of A to S .
- 5: $\bar{A} \leftarrow S/N$ ▷ Divide the sum by the number N of contacts.
- 6: **return**($\bar{A}$) ▷ Return the result, i.e. the average age of the N contacts.



Model

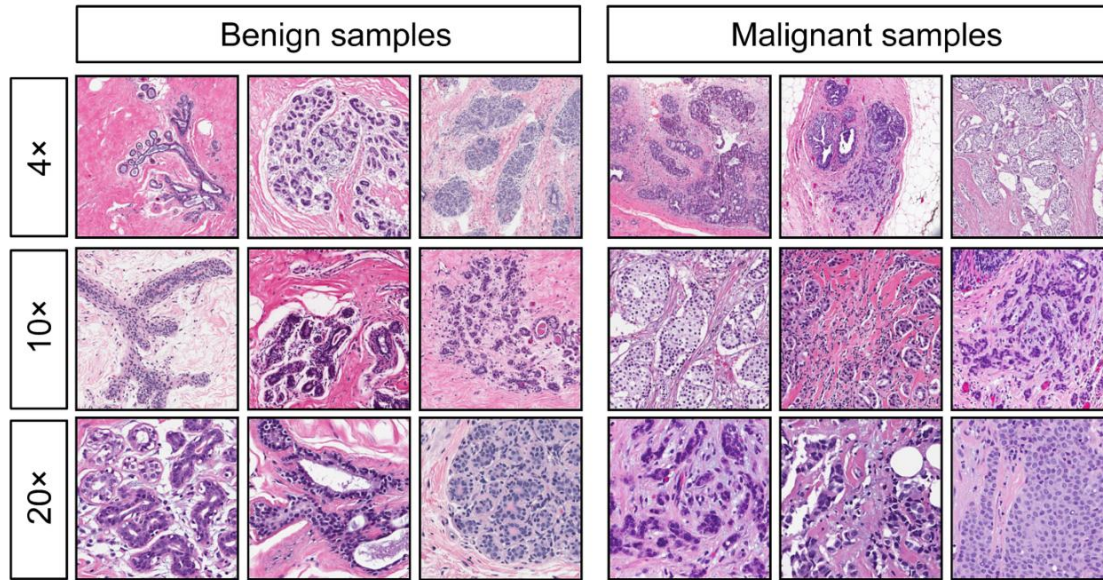
Example of Data Use

- Email classification



Example of Data Use

- Breast Cancer prediction



**Thank
You!**